

AMENDMENT NO. 4 SEPTEMBER 2026
TO
IS 16709 : 2017 TEXTILES — POLYPROPYLENE (PP) WOVEN, LAMINATED,
BLOCK BOTTOM VALVE SACKS FOR PACKAGING OF 50 KG CEMENT —
SPECIFICATION

(Page 3, clause 5.4, fourth sentence) — Substitute the following for the existing:

‘5.4 Lamination

If required by the buyer, the woven and laminated tubular fabric, before conversion into sacks, may be further laminated on the outer side with reverse-printed biaxially oriented polypropylene (BOPP) film having a minimum mass per unit area of 15 g/m².’

[Page 4, Table 1, Sl No. (iv)] — Substitute the following for the existing:

Sl No.	Characteristic	Requirement	Tolerance	Method of Test, Ref to
(1)	(2)	(3)	(4)	(5)
iv)	Mass of sack, g (<i>see</i> Note 2 and Note 3):			
	a) Without BOPP film	82	± 6 percent	Annex G
	b) With BOPP film	93		

(Page 4, Table 1, Note 3) — Substitute the following for the existing:

‘3 The mass of sack without BOPP film, as specified above, comprises of the mass of fabric (with 63 g/m²) and mass of lamination (minimum 23 g/m²) whereas the mass of sack with BOPP film, as specified above, comprises of the mass of fabric (with 63 g/m²), mass of lamination (minimum 23 g/m²) and mass of BOPP film (minimum 15 g/m²).’

(Page 11, Annex G) — Substitute the following for the existing:

‘ANNEX G

[Table 1, Sl No. (iv), Note 2]

TEST METHOD FOR DETERMINATION OF MASS OF SACK

G-1 PRINCIPLE

A finished woven sack as a test specimen is laid on a weighing balance in the folded form and the mass of the finished woven sack in grams is measured.

G-2 ATMOSPHERIC CONDITIONING AND TESTING

Prior to test, the specimens shall be conditioned to moisture equilibrium from dry side in the standard atmosphere of (65 ± 4) percent relative humidity and (27 ± 2) °C temperature or as laid down in IS 6359.

G-3 APPARATUS

G-3.1 Electronic Balance — capable of weighing test specimens with an accuracy of ± 0.1 g

G-4 PROCEDURE

G-4.1 A finished woven sack, as a test specimen, is placed on an electronic balance in the folded form.

G-4.2 Measure the mass of finished woven sack in grams with an accuracy of ± 0.1 g.

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G-4.3 Test at least 10 specimens and calculate the average mass of finished woven sack in grams.

G-5 RESULTS

The sample is found to be conforming if the average and each individual readings of mass of sack comply with the specified value of mass of sack, with or without BOPP film, including their applicable tolerances as mentioned in Table 1.

G-6 THEORETICAL CALCULATION FOR MASS OF SACKS (OPTIONAL)

G-6.1 Total mass of sacks comprises of:

- Mass of fabric;
- Mass of top and bottom patch;
- Mass of valve sleeve;
- Mass of lamination [see **G-6.2** {(d) + (e) + (f)}];
- Mass of BOPP film (If applicable) [see **G-6.2** {(g) + (h) + (j)}]; and
- Mass of printing ink, as per artwork.

G-6.2 Calculate the mass of sacks with the help of the following formula as the case may be:

- Mass of tubular fabric = $(a + c + 20) \times 2b \times M \times 10^{-6}$
- Mass of closure patch = $(d_b \times e_b + d_t \times e_t) \times M \times 10^{-6}$
- Mass of valve sleeve = $\{(v + 5) \times 2 \times i\} \times M \times 10^{-6}$
- Mass of lamination of tube = $\{(a + c + 20)\} \times 2(b + 4) \times M_1 \times 10^{-6}$
- Mass of lamination of patch = $2 \times (d_b \times e_b + d_t \times e_t) \times M_1 \times 10^{-6}$
- Mass of lamination of sleeve = $\{2 \times (v + 5) \times 2 \times i\} \times M_1 \times 10^{-6}$
- Mass of BOPP film of tube = $\{(a + c + 20)\} \times 2(b + 4) \times M_2 \times 10^{-6}$
- Mass of BOPP film of patch = $2 \times (d_b \times e_b + d_t \times e_t) \times M_2 \times 10^{-6}$
- Mass of BOPP film of sleeve = $\{2 \times (v + 5) \times 2 \times i\} \times M_2 \times 10^{-6}$
- Mass of printed ink = $A_p \times T_p \times P_p$

where

- a = sack length;
 c = closure height
 b = sack width
 M = mass of fabric, in g/m²;
 d_t = patch length (top closure);
 e_t = patch height (top closure);
 d_b = patch length (bottom closure);
 e_b = patch height (bottom closure);
 M_1 = mass, in g/m², of lamination, in g/m²;
 v = valve sleeve width;
 i = valve sleeve length;
 M_2 = mass, in g/m², of BOPP film, in g/m²;
 A_p = printed area, in m²;
 T_p = ink layer thickness, in micron; and

P_p = ink density, in g/cm³.

G-6.3 Test Reporting

G-6.3.1 Mass of sack without BOPP film, in g (*see* **G-6.2**) = [(a) + (b) + (c) + (d) + (e) + (f) + (k)].

G-6.3.2 Mass of sack with BOPP film, in g (*see* **G-6.2**) = [(a) + (b) + (c) + (d) + (e) + (f) + (g) + (h) + (j) + (k)].’